

- Q.28 Define brittleness elasticity, plasticity, Malleability.
Q.29 Discuss various grain defects found in wood.
Q.30 What are the benefits of using composite materials in aircraft.
Q.31 How is identification of non ferrous metal done by practical test?
Q.32 Explain various type of fabric and their specifications.
Q.33 What is the use of natural and synthetic rubber in aircrafts.
Q.34 Differentiate between iron and steel.
Q.35 Explain Case hardening process.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe various methods of Heat treatment in detail?
Q.37 Describe the various metals and alloys used in aircraft and standards if any in brief?
Q.38 Write a short note on any two-
a) Specification and use of Duralumin alclad.
b) Steel refining processes
c) Type of glue for aircraft use

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4th Semester/ Aircraft Maintenance Subject : Aircraft Materials & Material Science-I

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The tendency of a material to fracture without changing its shape is known as
a) Brittleness b) Hardness
c) Elasticity d) None of above
- Q.2 Which of the following materials is used in making aircraft windows?
a) Thick glass b) Plexi glass
c) Graphite d) Plane glass
- Q.3 The melting point of copper is?
a) 236 deg. Celsius b) 1085 deg. Celsius
c) 660 deg. Celsius d) None of above
- Q.4 Which heat treatment method is called drawing
a) Tempering b) Case hardening
c) Annealing d) Normalizing

Q.5 Which aluminium alloy is used for aircraft applications

- a) Duralumin b) Hindalium
- c) Magnelium d) Y alloy

Q.6 Copper is a _____ of electricity?

- a) Bad conductor b) Good conductor
- c) Typical insulator d) Medium conductor

Q.7 What happens to thermoplastics when they are heated?

- a) They become stronger
- b) They decompose
- c) They soften and become Moldable
- d) they become brittle

Q.8 Which iron is purest

- a) Pig Iron b) Wrought iron
- c) Cast Iron d) None of the above

Q.9 Which body issues Civil Airworthiness requirement in India

- a) FAA b) EASA
- c) DGCA d) ICAO

Q.10 What is the effect of heat treatment

- a) Improvement in availability
- b) Improved physical properties
- c) Decreased color
- d) Decreased availability

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

Q.11 What is the use of aircraft adhesives?

Q.12 Name any two type of Glue used in aircraft?

Q.13 Give an example of thermosetting plastic?

Q.14 Name any two type of Dopes.

Q.15 What is the use of plywood in aircraft?

Q.16 Name few steel types?

Q.17 Define Carburising.

Q.18 Name any two non ferrous metals?

Q.19 Name the ore of aluminium?

Q.20 What is purpose of heat treatment?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Describe in brief about aircraft adhesive and their uses.

Q.22 Explain different type of plastics.

Q.23 Write a short note on types of wood used in aircraft.

Q.24 Explain in brief various type of steel.

Q.25 Explain Ferrous and Non ferrous metals with example?

Q.26 What are the various non metallic material used in aircraft?

Q.27 State the factors effecting selection of materials for airplane parts?